Level 2



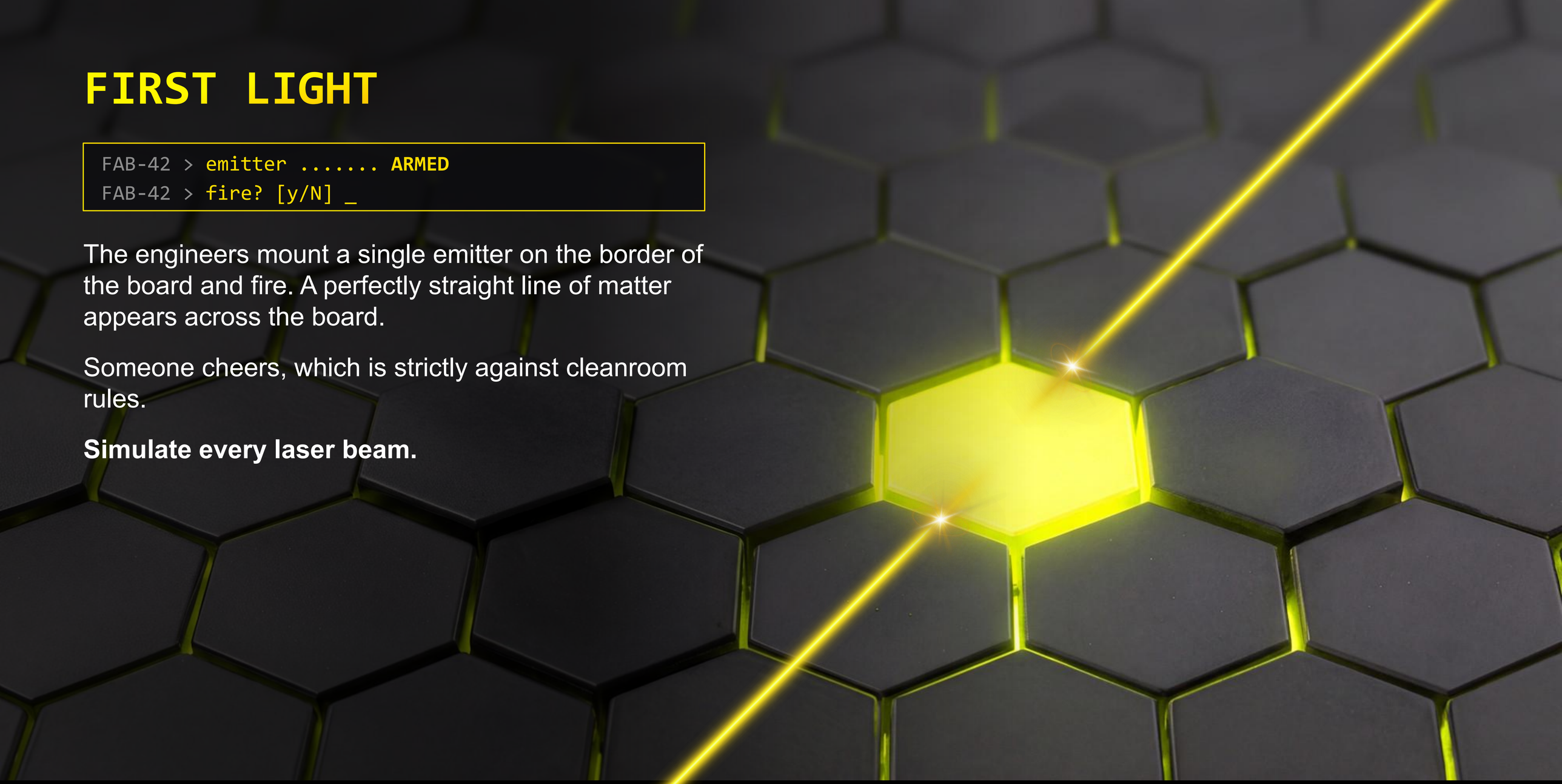
FIRST LIGHT

```
FAB-42 > emitter ....... ARMED
FAB-42 > fire? [y/N] _
```

The engineers mount a single emitter on the border of the board and fire. A perfectly straight line of matter appears across the board.

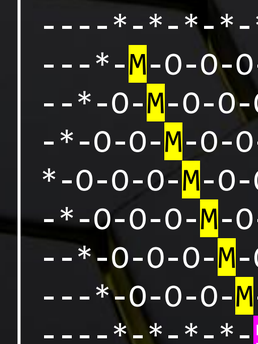
Someone cheers, which is strictly against cleanroom rules.

Simulate every laser beam.



- The nanofabricator has an emitter that produces a laser beam
- The emitter is placed on the border of the board
- The laser beam is a straight line that starts from the emitter
- Wherever the laser beam hits a cell of the board, matter is created
- The laser beam interacts only with ‘0’, not with ‘-’ or ‘*’
- The direction of the emitter is indicted by a number
  - Directions are explained on the next page
- You are given a list of boards with one emitter
- **For each board, determine the state after matter is created**

Emitter Directions

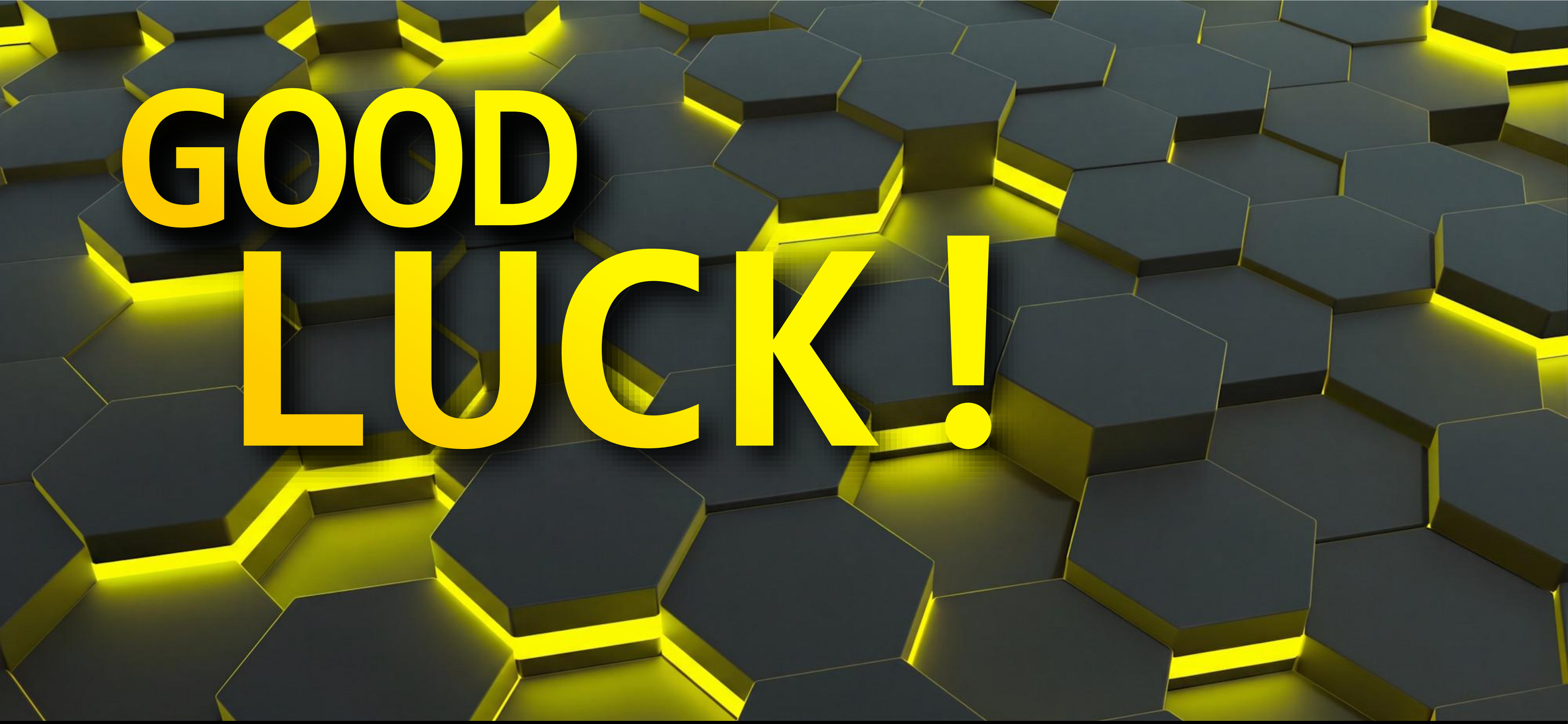
1 = →	2 = ↘	3 = ↙
		
4 = ←	**5 = ↖**	**6 = ↗**
		

Input

Name		Description	Example
N		Number of boards	6
repeated N times	Board dimensions	The radius of the board, the number of rows, and the number of columns, separated by spaces	4 9 17
	Board	A paragraph of characters ‘O’ represents an empty cell of the board ‘*’ represents the border of the board ‘-’ is only necessary for the indentation **‘1’-‘6’ represents the emitter**	----*-*-*-*-*---- ---*-0-0-0-0-*--- --*-0-0-0-0-0-*-- -*-0-0-0-0-0-0-*- 1-0-0-0-0-0-0-0-* -*-0-0-0-0-0-0-*- --*-0-0-0-0-0-*-- ---*-0-0-0-0-*--- ----*-*-*-*-*----
	Empty line between boards		4 9 17 ----*-*-*-*-*---- ---2-0-0-0-0-*--- --*-0-0-0-0-*-- ...

Output

Name		Description	Example
repeated N times	Board	A paragraph of characters ‘O’ represents an empty cell of the board **‘M’ represents matter** ‘*’ represents the border of the board ‘-’ is only necessary for the indentation *‘1’-‘6’ represents the emitter* *You can print the border with emitter or without. Both will be judged as correct.*	----*-*-*-*-*---- ---*-0-0-0-0-*--- --*-0-0-0-0-0-*-- -*-0-0-0-0-0-0-*- 1-M-M-M-M-M-M-M-* -*-0-0-0-0-0-0-*- --*-0-0-0-0-0-*-- ---*-0-0-0-0-*--- ----*-*-*-*-*----
	Mandatory empty line between boards		----*-*-*-*-*---- ---2-0-0-0-0-*--- --*-M-0-0-0-0-*-- ...



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